

AVAC4QGIS User Interface Guide

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1 General description of the interface

AVAC4QGIS is a dockable QGIS panel for preparing, running, and inspecting AVAC avalanche simulations. The user-selected Working Directory holds prepared runs, raw solver output, and QGIS result products. The packaged AVAC runtime is managed separately by the plugin.

The standard AVAC workflow has three tabs: *AVAC Parameters*, *AVAC Run*, and *Results*. The first contains both GIS inputs and scientific parameters. The usual sequence is: select the workspace and inputs, set or load parameters, validate and prepare an isolated run, run AVAC, then load results.

The optional Lake-Wave Extension adds *WAVE Parameters* and *WAVE Run* before the final *Results* tab. It also adds WAVE as a selectable result source inside *Results*. WAVE uses a completed AVAC run as a read-only source; it neither changes AVAC parameters nor overwrites AVAC output.

The parameter and result groups use accordion headers. Click a header to open that group. Status text and progress bars report the active operation, unavailable controls or eventual errors.

2 General simulation workflow

The following sequence describes a normal simulation and identifies where each step is performed in the interface.

1. Choose a writable **AVAC Working Directory** above the tabs. This is the location for the prepared runs and all derived result products.
2. In **AVAC Parameters**, select the terrain DEM and release polygons, then load a complete AVAC configuration or set the release, rheology, and numerical controls. The initial-depth and initial-snow-surface previews may be used here before a run is prepared.
3. In **AVAC Run**, use **Check Environment** and **Validate Inputs**, then **Prepare AVAC Run**. Preparation creates a self-contained data run below `runs/`; **Run** starts the packaged AVAC solver for that prepared run.
4. In **Results**, choose the completed AVAC run and load summary maps or time series. The same tab provides frame playback, PNG export, profiles, and gauges.
5. For a lake-impact case, enable the Lake-Wave Extension above the tabs after AVAC is complete. In **WAVE Parameters**, select the completed AVAC source, terrain/bathymetry, lake polygon, water level, calculation domain, and WAVE model controls. In **WAVE Run**, validate, prepare, and run the separate WAVE scenario. Return to the shared **Results** tab to load its maps, time series, profiles, gauges, and lake-volume history.

WAVE reads its selected AVAC run but never changes AVAC inputs or output. A change to an input or parameter after preparation makes that prepared run stale, so validate and prepare it again before running.

3 Function of each interface element

This section follows the visual order of the current interface. It deliberately does not document retired or hidden controls.

3.1 Controls above the tabs

3.1.1 AVAC Working Directory

This required field selects the writable user-data workspace. Type a path or use the `...` button to select or create a folder. It is not the plugin installation and it is not the packaged solver runtime. The plugin

creates `runs/` for AVAC simulations and `wave_runs/` for WAVE scenarios. QGIS remembers the selected path for the next session.

3.1.2 Enable Lake-Wave Extension

This check box makes the optional *WAVE Parameters* and *WAVE Run* tabs visible and adds WAVE to the final *Results* tab's source selector. Use it for a completed AVAC run that enters a lake or reservoir. Changing this check box does not modify any AVAC input, run, or result.

3.1.3 Help

The **Help** button, with the standard blue question-mark icon, opens this PDF. The same command is available in *Plugins > AVAC4QGIS*.

3.1.4 Complete Case Configuration

Save Case writes an *AVAC4QGIS plugin configuration* YAML file. It records the Working Directory, AVAC template and visible parameters, selected AVAC input layers, whether the Lake-Wave Extension is enabled, and, when enabled, all WAVE Parameters values and layer sources. **Load Case** restores that setup, including making the WAVE tabs visible when they were active at saving time.

3.2 AVAC Parameters tab

This tab contains the accordion groups *AVAC Inputs*, *Release / initial conditions*, *Rheology*, and *Simulation / numerical*.

3.2.1 AVAC Inputs

- **DEM raster layer** is required. It selects the AVAC terrain source. The plugin reads its first band. The raster must have a valid CRS, usable cells, and square cells. NoData values remain unavailable in the prepared terrain. Its cell size must equal, or be a multiple of the computational cell size.
- **Release polygon layer** is required. It must be a valid polygon or multipolygon vector layer. Every usable polygon feature defines initial avalanche area. The plugin transforms a preparation copy to the DEM CRS when necessary. At least one DEM grid point must occur within the combined release area.

For traceability, the plugin materializes local copies of selected sources in the prepared run folder.

3.2.2 Release / initial conditions

This group creates the snow-depth field at simulation time zero. The selected DEM is used unchanged as AVAC's fixed basal topography. The release polygon determines where the *mobile AVAC depth* is initially present; that depth is zero outside the selected release polygons. Consequently, the only snow state initialized by these controls is the mobile release thickness inside the release polygons.

- **Release depth** (d_0 , m) is the base initial snow thickness. With corrections off, every applicable grid point inside the release polygon receives this thickness. Increasing it increases initial modeled snow volume.
- **Critical slope** (θ_{cr} , degrees), **Reference elevation** (z_{ref} , m), **Hypsometric gradient**, and **Slope correction** ν supply the optional corrections.
- **Apply elevation correction** enables a linear depth adjustment. At z_{ref} , depth is d_0 ; the hypsometric gradient changes depth in meters per 100 m of elevation. A positive gradient makes initialized snow thicker above the reference elevation and thinner below it.

- **Apply slope correction** enables a factor derived from local DEM slope where the slope exceeds 25 degrees. It uses θ_{cr} and ν to modify the base depth and, when enabled, the elevation-corrected depth.
- **Return period [years]** is a positive whole-number scenario label saved with the prepared run and YAML configuration. It does not change the initial-depth calculation or the AVAC equations.

3.2.3 Rheology

This group selects the AVAC basal-resistance model and its material parameters.

- **Model** selects `Coulomb`, `Voellmy`, or `cohesive_Voellmy`. Coulomb uses a Coulomb basal-friction term. Voellmy combines it with speed-dependent resistance. Cohesive Voellmy adds a cohesive strength to the Voellmy law.
- **Density** (ρ , kg/m³) is the bulk snow density passed to AVAC. The same configured value is used when the plugin derives its maximum kinetic-pressure map.
- μ is the dimensionless Coulomb basal-friction coefficient. It is used by all three choices. Increasing it increases the Coulomb part of the resisting stress.
- ξ (m/s²) is the Voellmy speed-dependent resistance coefficient. It is enabled for `Voellmy` and `cohesive_Voellmy`, and disabled for `Coulomb`. In the implemented term, larger ξ produces less speed-dependent resistance.
- **Cohesion** (C , Pa) is enabled only for `cohesive_Voellmy`. It adds a material-strength term to that model's resisting stress, including its static yield condition.
- **Altitude zones** specifies bed-elevation-dependent rheology. Leave it blank for uniform parameters. Otherwise, enter one comma-separated row per zone:

μ , ξ , cohesion (Pa), lower elevation (m)

The first row has a blank final field; each later row has an ascending lower elevation. A zone applies from its lower limit to the next, and the last applies at and above the last break. For example, rows ending with a blank field and with 1680 define Zone 1 below 1680 m and Zone 2 at or above 1680 m.

- **Visualize Rheology Zones** creates and loads a categorical raster from the selected AVAC DEM. Every valid DEM cell is assigned the color of the altitude zone that the solver will use: cells at a lower elevation limit belong to the next, higher zone. The QGIS legend gives the elevation range and active rheology values for every color. The reusable preview file is saved under `qgis_previews` in the AVAC Working Directory; it does not change parameters or prepare a run.

3.2.4 Simulation / numerical

- **Simulation duration** is final modeled time in seconds.
- **AMR refinement levels** is the number of adaptive mesh refinement levels AVAC may use. More levels can resolve flow more finely but increase computation and output.
- **Output interval** is requested temporal-output spacing in seconds. The plugin derives the output count from it and duration so the final time is included. A smaller value creates more frames and output.
- **CFL target** and **CFL maximum** are numerical time-step limits. Target is the normal operating limit and cannot exceed maximum.

- **Computational cell size** (m) is base AVAC grid spacing. Smaller cells can represent terrain and flow variation more finely but cost more. It must equal the DEM cell size or be an integer multiple of it.
- **Boundary** selects `wall`, `extrap`, or `user`: `wall` prevents normal flow across the edge, `extrap` allows the solution to continue to the edge, and `user` uses configured AVAC boundary behavior.
- **Limiter** selects `none`, `minmod`, `superbee`, `MC`, or `van Leer` for numerical flux treatment. It is not a snow material parameter.

The plugin standardizes the non-user output settings for every prepared AVAC run: maximum-result checking at 1 s, solver verbosity 0, binary 32-bit output, and `Depth` as the default temporal variable. These settings are recorded in the prepared YAML but have no separate interface controls.

3.2.5 Load Configuration and Save Configuration

Load Configuration reads a complete AVAC YAML configuration, loads the visible values, and makes that file the template for preparation. **Save Configuration** writes a complete YAML file from that template and the current visible controls after validating parameter and grid rules.

3.3 AVAC Run tab

This tab validates, prepares, and runs the case selected on *AVAC Parameters*.

- **Active run** is read-only. After preparation it identifies the isolated run directory that **Run** executes.
- **CPU cores** sets the number of logical processor cores passed to AVAC as `OMP_NUM_THREADS`. It ranges from one to the maximum available on the current computer and defaults to that maximum. Reduce it when other work must remain responsive; the choice applies when **Run** starts.
- **Check Environment** checks the Working Directory, configuration template, and packaged runtime. It does not write a run or start AVAC.
- **Validate Inputs** checks selected DEM, release polygons, parameters, and configuration/grid compatibility without creating a run.
- **Prepare AVAC Run** makes a timestamped directory under `runs/`, materializes inputs, derives terrain and initial-condition files, writes configuration, and records metadata. It does not run AVAC. The preparation progress bar reports this operation.
- **Preview Initial Depth** creates a QGIS raster from the exact initial-depth computation used by preparation. Its reusable GeoTIFF is stored in `qgis_previews` inside the AVAC Working Directory. It runs in the background but does not prepare or execute AVAC.
- **Preview Initial Snow Surface** creates a second QGIS raster containing selected DEM elevation plus the exact mobile initial depth. This is the surface AVAC initializes at time zero. It is stored as `qgis_previews/avac_initial_snow_surface_elevation.tif`.
- **Run** starts the packaged AVAC solver for the prepared case. **Stop (direct process)** requests termination. An interrupted run remains on disk but is not a completed result.
- **Prepared Simulation**, status text, and the execution progress bar summarize readiness, messages, and solver progress.
- **Show execution log** reveals environment, validation, preparation, solver, and error messages.

3.4 WAVE Parameters tab

This tab appears only after enabling the Lake-Wave Extension. It prepares a separate GeoClaw WAVE scenario from a completed AVAC run.

3.4.1 AVAC Source

Completed AVAC run selects a completed run in the Working Directory. **Refresh Completed AVAC Runs** updates the choices. The selected AVAC frames supply WAVE coupling and are always read-only to WAVE.

3.4.2 Terrain and Lake Inputs

- **Terrain/bathymetry DEM** is WAVE terrain. It must cover the entire rectangular calculation domain and expected avalanche-entry area. It is typically the same DEM used for AVAC, but it can be a different raster with the same CRS. The plugin reads its first band and uses NoData values as unavailable terrain.
- **Lake water-body polygon** encloses the water body. It creates the force-dry mask and initial shoreline where the AVAC flow enters the lake. It is not the WAVE calculation domain. Land high enough above water level can become wet later through resolved wave runup.
- **Water level** (m) is absolute initial water-surface elevation in the same vertical datum as the terrain/bathymetry DEM. It is not water depth. It sets the initial water depth for all the DEM elements inside the Lake-Water body polygon over the water level.
- **Preview Water Level** makes a QGIS depth raster for wet cells inside the lake polygon. It evaluates only the selected calculation window at WAVE grid resolution and does not run the solver. Its reusable GeoTIFF is stored in `qgis_previews` inside the AVAC Working Directory.
- **Wave grid cell size** (m) must equal DEM resolution or be a whole-number larger multiple. A larger cell is coarser and reduces runtime and memory; a smaller cell is rejected because it requires unsupported silent interpolation.

3.4.3 WAVE Calculation Domain and Coupling Boundary

The WAVE solver runs in an axis-aligned rectangular domain specified by **X minimum**, **X maximum**, **Y minimum**, and **Y maximum** in the terrain DEM CRS. In this domain the plugin creates a WAVE grid, a dry mask outside the lake polygon, and internal faces along the initial shoreline for AVAC-to-WAVE coupling. The shoreline derived from the lake polygon is used to couple AVAC flow into WAVE. The rectangle must be large enough to contain the expected wave runup and the entire lake polygon.

- **Lake-domain buffer** (m) expands the lake-polygon extent when creating a domain automatically to have room for wave runup.
- **Create from Lake Polygon** creates the grid-aligned rectangle from the water-body extent and buffer.
- **Preview Calculation Domain** adds a temporary transparent polygon with red outline to QGIS for visual checking.

This rectangle is the solver domain, not the snow-to-water transfer line. During preparation, the plugin derives grid-aligned faces along the initial wet shoreline from the water-body polygon. It samples AVAC flow at these internal faces and creates WAVE internal forcing only for inward-moving flow that crosses into initially wet water. If no valid inward crossing is found, preparation reports that the lake was not reached and leaves WAVE unavailable to run.

3.4.4 WAVE Model Settings

- **Snow-to-water damping** is a multiplier on the internally coupled AVAC-to-WAVE source rates. It is used to transition from the AVAC snow model to the WAVE water model. A value of 1.0 means no damping; smaller values reduce the source term and produce a weaker wave.

- **Land Strickler coefficient** and **Water Strickler coefficient** set WAVE roughness on land and in water. The solver uses their reciprocals as Manning coefficients, so a larger Strickler value means weaker Manning friction.
- **Friction depth limit** (m) sets the depth above which the WAVE friction source term is not applied.
- **Dry-depth limit** (m) is the threshold for WAVE wet/dry handling.
- **Wave refinement tolerance** controls the WAVE refinement flag. **CFL target** and **CFL maximum** are numerical time-step controls and must satisfy $\text{target} \leq \text{maximum}$.
- **Limiters** selects none, minmod, superbee, MC, or van Leer for WAVE numerical flux treatment.

Save WAVE Configuration writes a portable *AVAC4QGIS WAVE setup configuration* YAML file containing all WAVE Parameters values: terrain and lake-polygon layer sources, water level, grid and calculation-domain settings, model settings, and the selected completed AVAC source run. **Load WAVE Configuration** restores these controls. If the recorded completed AVAC run is not present in the current Working Directory, the WAVE setup is still loaded, but a completed AVAC source must be selected before preparation. This WAVE-only file does not change AVAC inputs or parameters.

3.5 WAVE Run tab

- **Prepared Simulation** reports the active prepared WAVE scenario. Before preparation it states the required sequence.
- **CPU cores** sets the number of logical processor cores passed to the WAVE solver through its OpenMP thread setting. It defaults to the maximum available on the current computer; choose a smaller value to reserve CPU capacity for QGIS or other applications.
- **Check Environment** verifies the Working Directory and bundled WAVE runtime. **Validate Inputs** checks the completed AVAC source, terrain, lake polygon, calculation domain, cell size, and CRS without creating a scenario.
- **Prepare Wave Run** creates an isolated scenario in `wave_runs/`, stages terrain and masks, creates shoreline-coupling data, and inherits the source AVAC duration and output cadence. The prepared lake state from **Preview Water Level** is reused when unchanged; if no current preview exists, preparation calculates it once.
- **Run** is available after successful WAVE preparation with detected lake inflow. It starts the packaged WAVE solver in its separate scenario directory.
- **Stop** requests a stop. If the solver remains active after a grace period, the plugin force-stops it. An interrupted scenario remains on disk but is not a completed result.
- Status text and the progress bar report state and count WAVE output frames. **Show Wave execution log** reveals preparation, runtime, solver, and error messages.

3.6 Results tab

Results is always the last workflow tab. AVAC and WAVE are not separate result pages: the same Summary Maps, Time Series Maps, Profile Analysis, and Gauge Histories controls operate on the solver named in the selected product. When the Lake-Wave Extension is enabled, WAVE run selection, WAVE variables, and Lake Volume appear in those shared controls. The progress bar at the bottom reports result loading and export progress.

3.6.1 Results Run

Completed workspace run, **Refresh Runs**, and **Open Run Directory...** select a completed AVAC run. When WAVE is enabled, **Completed WAVE scenario**, its Refresh Runs button, and **Open Wave Scenario Directory...** select a completed WAVE scenario in the same group. Both summaries remain visible, so an AVAC run and a related WAVE scenario can be loaded together.

3.6.2 Summary Maps

Summary map selects Maximum Depth, Maximum Velocity, Maximum Pressure, Arrival Time, Time of Maximum Depth, or Time of Maximum Velocity. These are AVAC products. When WAVE is enabled, the same selector also offers WAVE Maximum surface rise and Maximum surface drawdown relative to the initial water surface. **Load Map** dispatches to the selected solver.

3.6.3 Time Series Maps and PNG export

- **Variable** identifies both solver and quantity. AVAC offers Depth, Velocity, Pressure, and Snow Surface Elevation. When WAVE is enabled, **WAVE — Snow Depth Outside Lake** is also available for visualization purposes. It is a separate copy of AVAC Depth in which AVAC grid cells intersecting the prepared WAVE lake footprint are set to zero for every frame; the original AVAC Depth product remains unchanged. It preserves AVAC's band count and elapsed simulation-time axis, so the shared Frame Player plays and seeks it exactly as it does the AVAC time series. Select the completed WAVE scenario prepared from the selected AVAC run before loading it. WAVE adds Water depth, Water elevation, and Surface displacement. Water elevation is the modeled free-surface elevation in the terrain DEM vertical datum; dry cells are NoData. Surface displacement is change from the initial free surface.
- **Load Time Series** creates or reuses the selected multiband GeoTIFF and loads it into QGIS. Loaded AVAC and WAVE series can remain visible together. Profile extraction, gauge sampling, and selected-map or frame export load the requested time series automatically when it is not already loaded.
- The built-in **Frame Player** provides play, pause, restart, and direct seeking.
- **Extent** applies only to PNG rendering: current map canvas or selected result extent. It does not change a calculation domain or raster.
- **Image width** sets 640–8000 px. **Simulation time** controls the elapsed-time title. **Legend** adds the color bar of every visible AVAC or WAVE temporal raster, with separate bars when their ranges differ. **Scale bar** adds map distance at lower left.
- **Export Current Map PNG** writes the displayed map or frame. **Export Time Series Frames...** writes the selected temporal sequence, while **Export every Nth frame** controls sampling. **Cancel Export** stops the sequence; files already written remain.

3.6.4 Profile Analysis

- **Profile layer** lists QGIS line layers; **Refresh Layers** updates the list. A one-feature layer is used automatically. With multiple lines in the same layer, select exactly one feature in QGIS using the

"Select Features by Area or Single Click" tool.

- **Result type** separates three temporal interpretations of the chosen **Variable**: **Selected frame** samples the currently displayed simulation frame; **Full time series (export)** prepares a history export while retaining the selected frame for a preview; and **Historical maximum through selected frame** samples the largest value reached at every profile point from time zero through the currently displayed frame.
- **Variable** identifies an AVAC snow quantity, the derived WAVE Snow Depth Outside Lake, or a WAVE water quantity when the extension is enabled.
- **Sampling** chooses raster-cell spacing or user-defined spacing. **Extract / Plot Profile** samples and opens the profile; **Export CSV** saves the selected result type.
- For **Full time series (export)**, **Time-series export** can write a long-form CSV or one PNG profile per AVAC or WAVE frame.

3.6.5 Gauge Histories

Gauge histories are sampled from the selected AVAC or WAVE time series.

- **Point layer** accepts an optional QGIS point layer.
- **Variable** independently chooses AVAC Depth, Velocity, Pressure, or Snow Surface Elevation and, when WAVE is enabled, WAVE Water depth, Water elevation, or Surface displacement.
- **Read Gauges from Selected Point Layer** automatically loads and materializes the selected time series when it is not already in the project, then samples every usable point from every band with CRS transformation when needed.
- **Sampled point** selects a one point feature from the chosen layer.
- **Refresh Points** updates the list of available points.
- **Plot Gauge History** opens it and **Export Gauge CSV** saves it.

3.6.6 Lake Volume

Lake volume is computed at every output time as modeled depth times cell area, summed over the initial water-body mask. **Plot Lake Volume History** and **Export Lake Volume CSV** display or save it.

4 Working-directory structure

The Working Directory contains user data. A normal AVAC run is arranged as:

```
AVAC Working Directory/  
  runs/  
    run_YYYYMMDD_HHMMSS/  
      .avac_qgis_run.json  
      inputs/                materialized input data and source records  
      Topo/                  AVAC terrain inputs  
      AVAC/                  configuration, initial condition, and _output/  
      qgis_results/          derived QGIS rasters and results.json
```

Run directory names and displayed timestamps use the computer's local time. AVAC and WAVE scientific output times remain elapsed simulation seconds.

WAVE scenarios are siblings of AVAC runs:

```

AVAC Working Directory/
  wave_runs/
    wave_YYYYMMDD_HHMMSS/
      .avac_qgis_wave.json
      Topo/                          WAVE terrain and mask inputs
      CL/                            internal shoreline coupling data
      Wave/_output/                  raw WAVE solver output
      qgis_wave_results/            derived WAVE rasters and diagnostics

```

The WAVE directory records a reference to its AVAC source run. It does not copy, overwrite, or delete AVAC source output.